Hengameh Mirhajianmoghadam

Robotics Researcher and Engineer | Control, State Estimation and Perception

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Professional Summary

Robotics researcher and engineer with doctoral research in Electrical and Computer Engineering and industry experience in embedded control and motion systems. Develops model predictive control, nonlinear control, system identification, and multi-agent coordination, alongside event-based perception, sensor fusion, and biologically inspired localization and mapping. Combines algorithm development in C++ and Python with ROS integration and experimental evaluation on aerial and ground robots. Experience also spans deep learning for visual detection, parallel-robot control, university teaching, and graduate and undergraduate research mentoring.

Technical Expertise

Control and optimization: Linear and nonlinear MPC, decentralized control, PID, LQR, trajectory tracking, system identification, dynamics modeling, SQP, CasADi, do-mpc, and Ipopt.

State estimation and perception: Kalman filtering, camera–IMU fusion, event-based vision, visual odometry, biologically inspired SLAM, place recognition, loop closure, LiDAR processing, and CNN-based object detection.

Programming and software: C++, Python, MATLAB/Simulink, Bash, Linux/Ubuntu, Git, OpenCV, PyTorch, CUDA, ROS, ROS 2 Humble, Gazebo, and basic Docker.

Hardware and validation: Parrot Bebop UAS, Quanser QCar2, NVIDIA Jetson AGX Orin, event cameras, IMUs, OptiTrack, Raspberry Pi, motor drives, position sensors, Kalibr, sensor calibration, timestamp synchronization, and rosbag replay.

Research and Industry Experience

Graduate Research Assistant — Autonomous Systems, Robotics and Control

New Mexico State University, Unmanned Systems Laboratory | Jan 2022–Present

- **Real-time drone control.** Identified commercial quadrotor dynamics and control delay from flight data using extended least squares; developed and experimentally evaluated constrained MPC against PID for fast trajectory tracking.
- Implemented the controller in ROS with Python, do-mpc, and CasADi at a 100 Hz closed-loop rate, with optimization times below 10 ms per iteration; evaluated tracking against OptiTrack motion-capture feedback.
- **Multi-agent coordination.** Co-developed sigma-norm decentralized MPC for lattice formation and collision-aware flocking using local information; compared performance with three established methods in simulation.
- **Event-based localization and mapping.** Developed eRatSLAM components combining event-aligned motion estimation, Kalman filtering with IMU measurements, time-surface place recognition, and biologically inspired cognitive mapping.
- **Embedded perception and control.** Implemented event-camera lane detection directly from events on QCar2 with Jetson AGX Orin and ROS 2 Humble, connecting lane perception with PID and MPC vehicle control.

Research and Industry Experience Continued

Researcher — Computer Vision and Deep Learning

Ferdowsi University of Mashhad, Machine Vision Laboratory | Feb 2020–Nov 2021

- Developed EYNet for airport detection in remote-sensing images, extending YOLOv3 with complementary feature-extraction networks and Shearlet-based visual attention.
- Implemented preprocessing, data augmentation, negative mining, and GPU training using Python, PyTorch, OpenCV, and CUDA; evaluated detection across varied scales and complex backgrounds, resulting in a first-author journal publication.

Research and Development Engineer — Embedded Control

Engarium Planetarium, Ferdowsi University of Mashhad | Jan 2018–Oct 2019

- Developed an automated observatory telescope-dome positioning system by integrating motor drives, position sensing, and embedded feedback control.
- Designed and installed the control system, including PID positioning and safety interlocks, to support regular observatory operation.

Research and Development Engineer — Motion Systems

CanDo VR Company | May 2016–Nov 2017

- Developed a pneumatic motion platform using Raspberry Pi and Unity to synchronize physical motion with a virtual-reality simulation under changing rider loads.
- Carried the system from prototype development to a commercial product used in an entertainment park.

Graduate Researcher — Nonlinear Control and Intelligent Systems

Ferdowsi University of Mashhad, Center of Excellence on Soft Computing | Sep 2013–Feb 2016

- Developed nonlinear MPC for position tracking and torque control of a 3-PSP parallel robot using MATLAB, sequential quadratic programming, and a hierarchical prediction structure.
- Developed differentiable harmonic emotional neural networks for nonlinear system identification and predictive control; improved tracking and reduced average computation time approximately twenty-fold compared with hierarchical MLP-based MPC in simulation.
- Evaluated robustness to measurement noise, disturbances, and parameter uncertainty; the research produced journal and conference publications.

Education

Ph.D. in Electrical and Computer Engineering

New Mexico State University, Las Cruces, New Mexico | Jan 2022–Dec 2026

Research: model predictive control, event-based perception and state estimation, biologically inspired localization and mapping, and autonomous robotic systems.

M.Sc. in Electrical and Computer Engineering

Ferdowsi University of Mashhad, Mashhad, Iran | Sep 2013–Feb 2016

Research: emotional neural networks for nonlinear system identification and model predictive control.

Teaching and Mentoring

Primary Instructor — EE 407 Introduction to Control Systems

New Mexico State University | Fall 2025

- Independently planned and delivered lectures, assignments, examinations, MATLAB/Simulink activities, and student support; taught dynamic-system modeling, stability, root locus, frequency response, and feedback-controller design.

Teaching Assistant and Laboratory Developer — EE 475/551

New Mexico State University, Control Systems Synthesis | Spring 2026

- Developed MATLAB/Simulink and Quanser QLabs digital-twin laboratory modules for transfer-function identification, model validation, robustness analysis, and LQR regulation and tracking in a combined senior undergraduate and graduate course.

Additional teaching assistantships — New Mexico State University

EE 362 Introduction to Computer Organization — Spring 2026
ME 210 Electronics and System Engineering — Fall 2024; Spring 2025
EE 325 Signals and Systems II — Fall 2023; Spring 2024
EE 407 Introduction to Control Systems — Fall 2023

Research and project mentoring

- Mentored graduate and undergraduate students in event-based perception, SLAM, robotic control, software implementation, experimental design, technical documentation, and research presentations.
- Mentored graduate students on QCar2 LiDAR navigation, occupancy-grid mapping, and A* / D* Lite path planning; supported undergraduate capstone implementation and troubleshooting.

Honors and Awards

Student Travel Support Award, IEEE CCTA — 2025
RAS Travel Support Award, IEEE CASE — 2025
Merit-based Enhancement Award, NMSU — Fall 2024; Spring 2025
Engineering Research Award, NMSU — Spring 2024
Graduate Success Scholarship, NMSU — Fall 2022; Spring 2023

Leadership and Professional Service

Session Chair, IEEE Conference on Control Technology and Applications — 2025
Judge, Undergraduate Research and Creative Arts Symposium, NMSU — 2025
President, Graduate Student Association, NMSU — 2023–2024

Peer review — IEEE Transactions on Aerospace and Electronic Systems (2025–2026); Frontiers in Robotics and AI (2022); American Control Conference (2027 edition, review completed 2026); ICARCV (2024); ICRA (delegate reviewer, 2024); CDC (2023); IFSA–EUSFLAT (subreviewer, 2021).

Professional Certificates

DeepLearning.AI — Neural Networks and Deep Learning; Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization; Structuring Machine Learning Projects.

Publications

Peer Reviewed Journal Articles

[1] E. E. Hernández, **H. Mirhajianmoghadam**, E. S. Espinoza, and L. R. Garcia Carrillo. "Predictive Flocking of an Omnidirectional Multi-Agent System for Lattice Formation: Sigma-Norm Design and Comparative Performance Assessment" *IEEE Access, vol. 14, pp. 3910–3925, 2026*.

[2] **H. Mirhajianmoghadam** and B. Bolourian Haghighi. "Eynet: Extended YOLO for Airport Detection in Remote Sensing Images" *Pattern Analysis and Applications, vol. 29, article 10, 2026*.

[3] **H. Mirhajianmoghadam** and M.-R. Akbarzadeh-T.. "Predictive Hierarchical Harmonic Emotional Neuro-Cognitive Control of Nonlinear Systems" *Engineering Applications of Artificial Intelligence, vol. 111, article 104781, 2022*.

Peer Reviewed Conference Papers

[4] **H. Mirhajianmoghadam** and L. R. Garcia Carrillo. "Bringing Loop Closure to Event-Based SLAM: A Place Recognition Approach" *IEEE International Conference on Automation Science and Engineering (CASE), pp. 2124–2129, 2025*.

[5] **H. Mirhajianmoghadam** and L. R. Garcia Carrillo. "EVO-RatSLAM: A Biologically-Inspired SLAM for Event-Based Visual Odometry" *IEEE Conference on Control Technology and Applications (CCTA), pp. 483–488, 2025*.

[6] **H. Mirhajianmoghadam** and L. R. Garcia Carrillo. "Event-Based RatSLAM: Enhancing RatSLAM with Event-Aligned Velocity Estimation and Kalman Filtering for Mapping" *IEEE International Conference on Artificial Intelligence, Robotics and Control (AIRC), pp. 164–168, 2025*.

[7] **H. Mirhajianmoghadam**, N. Grijalva, and L. R. Garcia Carrillo. "Fast Trajectory Tracking for Commercial Off-the-Shelf UAS Using Model Predictive Control: Real-Time Implementation and Performance Analysis" *International Conference on Unmanned Aircraft Systems (ICUAS), pp. 658–665, 2024*.

[8] N. Grijalva, **H. Mirhajianmoghadam**, L. R. Garcia Carrillo, H. Stuckey, and W. Tang. "Low-Cost Monocular Vision-Based Localization and Feedback Control of UAS for Classroom and Education Settings" *International Conference on Unmanned Aircraft Systems (ICUAS), pp. 667–672, 2024*.

[9] **H. Mirhajianmoghadam**, N. Grijalva, and L. R. Garcia Carrillo. "Real-Time Implementation of Model Predictive Control for Fast Trajectory Tracking of a Quad Rotorcraft UAS" *IEEE International Conference on Robotic Computing (IRC), pp. 90–91, 2023*.

[10] **H. Mirhajianmoghadam**, M.-R. Akbarzadeh-T., and E. Lotfi. "A Harmonic Emotional Neural Network for Non-Linear System Identification" *24th Iranian Conference on Electrical Engineering (ICEE), 2016*.

Manuscripts Under Review

[11] **H. Mirhajianmoghadam** and L. R. Garcia Carrillo. "eRatSLAM: Event-Based SLAM and Perception for Autonomous Robotic Systems" *Manuscript under review, 2026*.

[12] H. C. Contreras Aguirre, **H. Mirhajianmoghadam**, O. A. Garcia Alcantara, G. Contreras, and L. R. Garcia Carrillo. "Experiential Learning in Control Systems Education Through Robotics and Digital-Twins" *Manuscript under review, 2026*.

Preprint

[13] **H. Mirhajianmoghadam** and S. M. Ghasemi. "Efficient Parameter Selection for Scaled Trust-Region Newton Algorithm in Solving Bound-Constrained Nonlinear Systems" *arXiv:2009.04354, 2020*.